IETF-125 Hackathon

I2ICF Side Meeting

Interface to In-Network Computing Functions (I2ICF) Project

March 15, 2026, Shenzhen

Champion: Jaehoon (Paul) Jeong

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IETF-125 Interface to In-Network Computing Functions (I2ICF)

Champion: Jaehoon Paul Jeong (SKKU)

IETF-125 Interface to In-Network Computing Functions (I2ICF) Hackathon

Professors:

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- Joo-Sang Youn (DEU)

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- Kihyun Shim (SKKU)
- Deokhyeon Ryu (DJU)
- Eunsan Seong (DEU)
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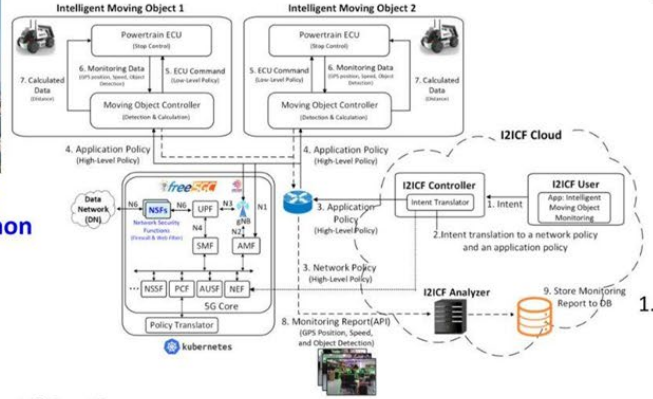
I2ICF Framework



Objectives

- To demonstrate an intent-driven, closed-loop safety pipeline in which a moving object receives an intent, this I2ICF framework (i) performs on-board camera-based object detection and camera–LiDAR calibration/fusion for distance estimation, (ii) streams detections to a cloud server, and (iii) executes real-time stop/detour for obstacles and pedestrians.

Future Work

- To optimize End-to-End performance (e.g., inference accuracy, and inference latency) and support two robot cars so that they exchange on-board inference data to cooperatively plan collision-free, optimal paths when approaching each other.

I2ICF Developing Environment

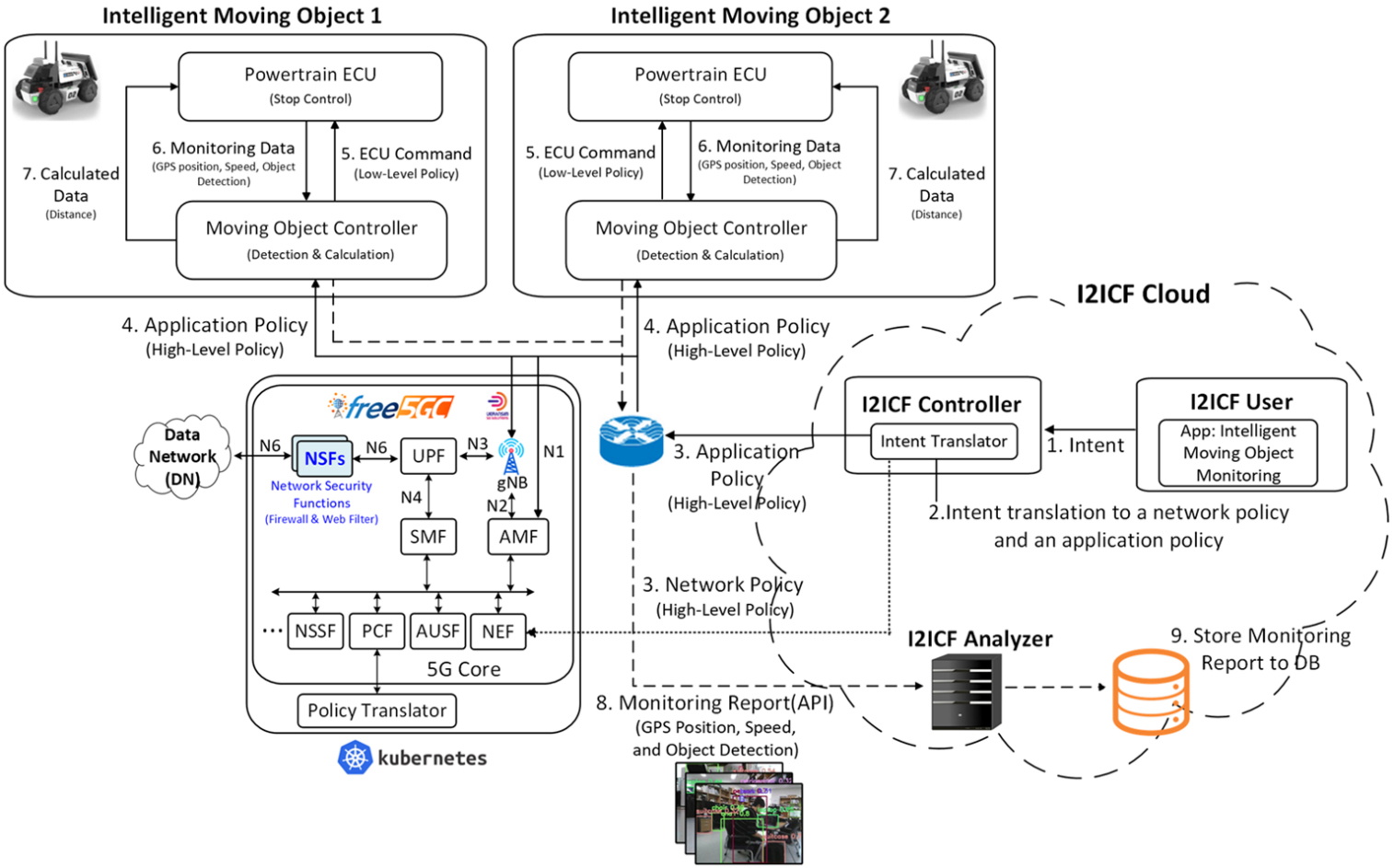
- OS: Ubuntu 20.04
- Kubernetes: Microk8s v1.32.2
- Object Detection: YOLOv8
- ROS2 Version: Humble
- GitHub Repository: https://github.com/jaehoonpauljeong/I2ICF/tree/main/IETF-125

Workflow of the I2ICF Testbed

1. Intent Submission: A User provides a safety/perception intent (e.g., avoid obstacles/pedestrians and stop within a configured distance).
2. Intent Application: A Moving Object sets detection/stop parameters from the intent.
3. Calibration: Camera–LiDAR calibration and synchronization for distance estimation.
4. Perception & Fusion: Camera detection and LiDAR range fusion to classify objects and estimate distance.
5. Cloud Streaming: Detection results and motion metadata are sent to a cloud server (I2ICF Analyzer).
6. Safety Action: If a proximity threshold is met, the Moving Object stops to avoid a collision and detours for a safe driving.
7. Monitoring: Logs (e.g., JSON) are stored in the cloud database (DB) for further analysis.

성균관대학교 SUNGKYUNKWAN UNIVERSITY | 대전대학교 DAEJEON UNIVERSITY | 동의대학교 DONG-EUI UNIVERSITY | ETRI 한국전자통신연구원 | KETI Korea Electronics Technology Institute | TTA 한국정보통신기술협회 Telecommunications Technology Association

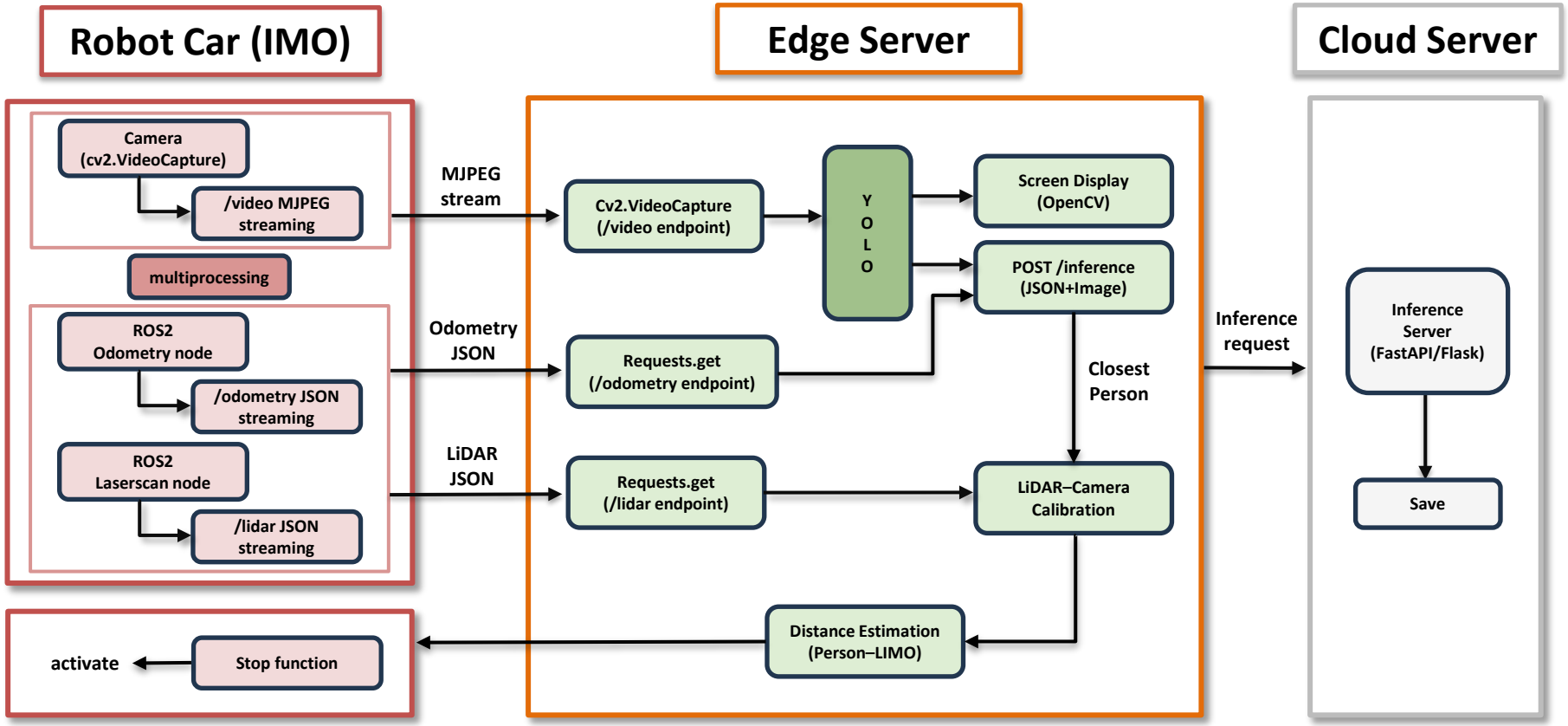
Interface to In-Network Computing Functions (I2ICF) for Mobile Objects



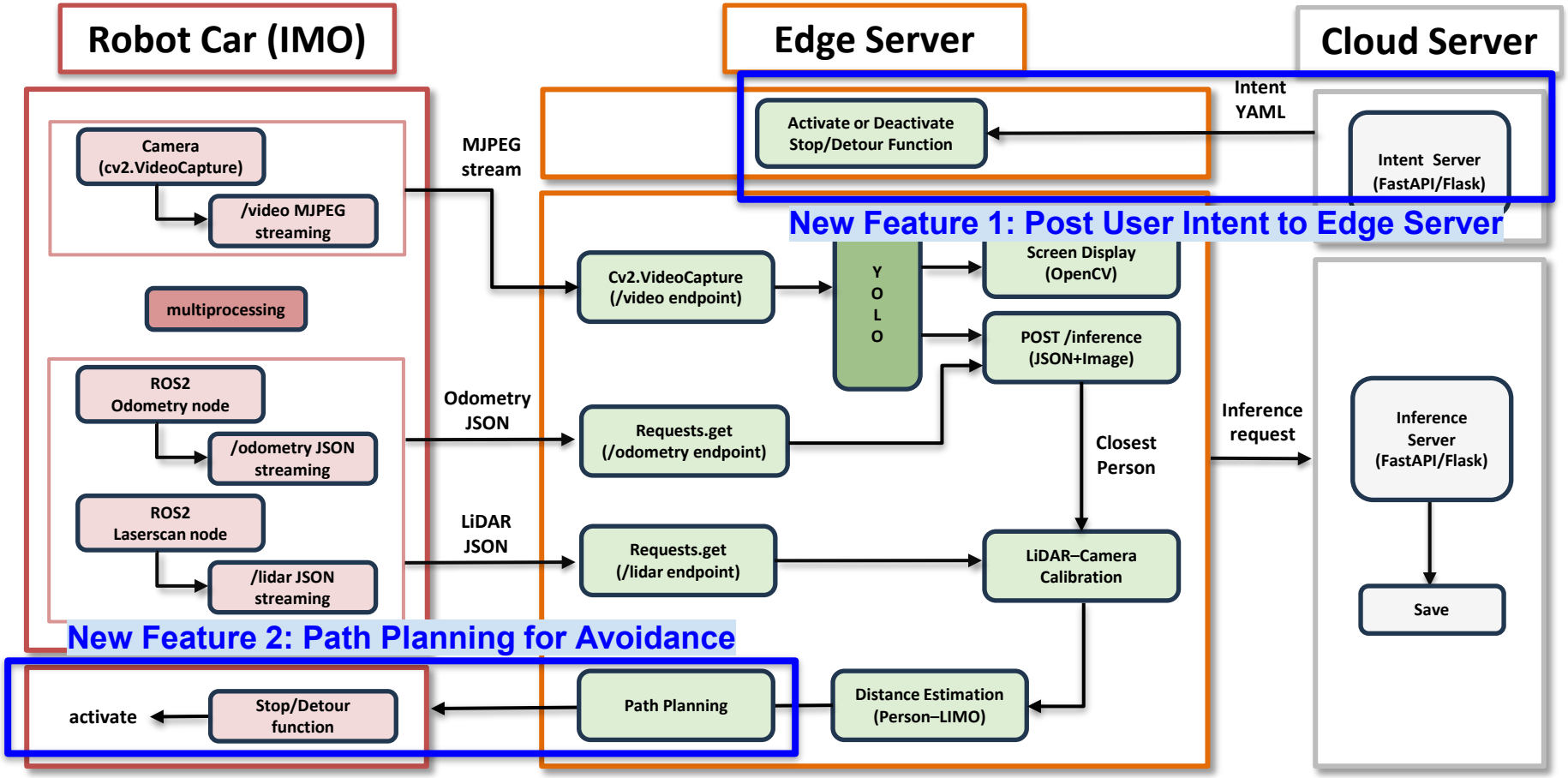
Goal of Hackathon Project

- The goal is to show the feasibility of Configured Moving Objects to Collision Avoidance by self-determining obstacles and pedestrians
  - **I2ICF (Interface to In-Computing Functions) Moving Object Controller**
    - The robot car fuses the sensing data of both camera and LiDAR in real time to measure a distance to an object.
    - The robot car automatically detours for collision avoidance when a detected person is too close to it.
- Internet Drafts for the I2ICF Project
  - https://datatracker.ietf.org/doc/draft-jeong-opsawg-i2icf-problem-statement/
  - https://datatracker.ietf.org/doc/draft-jeong-opsawg-i2icf-framework/
  - https://datatracker.ietf.org/doc/draft-gu-nmrg-intent-translator/

IETF-124 I2ICF Framework (Before)



IETF-125 I2ICF Framework (Now)



Demonstration (1/4)

```
agilex@agikex-NUC12...   agilex@agikex-NUC12...   agilex@agikex-NUC12...   agilex@agikex-NUC12...
agilex@agikex-NUC12WSKi7:~$ ros2 launch limo_base limo_base.launch.py
[INFO] [launch]: All log files can be found below /home/agilex/.ros/log/2026-03-10-17-12-02-475675-agike
x-NUC12WSKi7-14123
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [limo_base-1]: process started with pid [14124]
[limo_base-1] Loading parameters:
[limo_base-1] - port name: ttylimo
[limo_base-1] - odom frame name: odom
[limo_base-1] - base frame name: base_link
[limo_base-1] - pub odom tf: 0
[limo_base-1] -use_mcnamu: 0
[limo_base-1] [INFO] [1773133922.519878099] [limo_base_node]: connet the serial port:'/dev/ttylimo'
[limo_base-1] [INFO] [1773133922.521350853] [limo_base_node]: enableCommandedMode :
[limo_base-1] [INFO] [1773133922.521359403] [limo_base_node]: Open the serial port:'/dev/ttylimo'
```

1. [LIMO] Run ‘limo_base.launch.py’

```
agilex@agikex-NUC12...   agilex@agikex-NUC12...   agilex@agikex-NUC12...   agilex@agikex-NUC12...
agilex@agikex-NUC12WSKi7:~$ ros2 launch ydlidar_ros2_driver ydlidar.launch.py
[INFO] [launch]: All log files can be found below /home/agilex/.ros/log/2026-03-10-17-12-09-077565-agike
x-NUC12WSKi7-14137
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [ydlidar_ros2_driver_node-1]: process started with pid [14148]
[INFO] [static_transform_publisher-2]: process started with pid [14150]
[static_transform_publisher-2] [WARN] [1773133929.139142902] []: Old-style arguments are deprecated; see
 --help for new-style arguments
[ydlidar_ros2_driver_node-1] [INFO] [1773133929.146685493] [ydlidar_ros2_driver_node]: [YDLIDAR INFO] Cu
rrent ROS Driver Version: 1.0.1
[ydlidar_ros2_driver_node-1]
[ydlidar_ros2_driver_node-1] YDLidar SDK initializing
[ydlidar_ros2_driver_node-1] YDLidar SDK has been initialized
[ydlidar_ros2_driver_node-1] [YDLIDAR]:SDK Version: 1.1.3
[static_transform_publisher-2] [INFO] [1773133929.147912911] [static_tf_pub_laser]: Spinning until stopp
ed - publishing transform
[static_transform_publisher-2] translation: ('0.000000', '0.000000', '0.020000')
[static_transform_publisher-2] rotation: ('0.000000', '0.000000', '0.000000', '1.000000')
[static_transform_publisher-2] from 'base_link' to 'laser_link'
[ydlidar_ros2_driver_node-1] LiDAR successfully connected
[ydlidar_ros2_driver_node-1] [YDLIDAR]:Lidar running correctly ! The health status: good
[ydlidar_ros2_driver_node-1] [YDLIDAR INFO] Current SDK does not support current lidar models[unkown(YD-
151)]
[ydlidar_ros2_driver_node-1] LiDAR init success, Elapsed time 766 ms
[ydlidar_ros2_driver_node-1] Start to getting intensity flag
[ydlidar_ros2_driver_node-1] lastPos 98 currPos 111 offset 13
[ydlidar_ros2_driver_node-1] Auto set intensity 1
```

2. [LIMO] Run ‘ydlidar.launch.py’

```
agilex@agikex-NUC12...   agilex@agikex-NUC12...   agilex@agikex-NUC12...   agilex@agikex-NUC12...
agilex@agikex-NUC12WSKi7:~$ python3 /home/agilex/agilex_ws/src/SDV_Robocar/imo_server_lidar.py
[INFO] Flask + ROS2 병렬 실행 시작
 * Serving Flask app 'imo_server_lidar'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI s
erver instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:8000
 * Running on http://192.168.50.82:8000
Press CTRL+C to quit
192.168.50.231 - - [10/Mar/2026 17:12:25] "GET /odometry HTTP/1.1" 200 -
192.168.50.231 - - [10/Mar/2026 17:12:25] "GET /lidar HTTP/1.1" 200 -
192.168.50.231 - - [10/Mar/2026 17:12:25] "GET /odometry HTTP/1.1" 200 -
192.168.50.231 - - [10/Mar/2026 17:12:25] "GET /lidar HTTP/1.1" 200 -
192.168.50.231 - - [10/Mar/2026 17:12:25] "GET /odometry HTTP/1.1" 200 -
192.168.50.231 - - [10/Mar/2026 17:12:25] "GET /odometry HTTP/1.1" 200 -
192.168.50.231 - - [10/Mar/2026 17:12:25] "GET /lidar HTTP/1.1" 200 -
192.168.50.231 - - [10/Mar/2026 17:12:25] "GET /odometry HTTP/1.1" 200 -
192.168.50.231 - - [10/Mar/2026 17:12:25] "GET /lidar HTTP/1.1" 200 -
192.168.50.231 - - [10/Mar/2026 17:12:25] "GET /odometry HTTP/1.1" 200 -
192.168.50.231 - - [10/Mar/2026 17:12:26] "GET /odometry HTTP/1.1" 200 -
192.168.50.231 - - [10/Mar/2026 17:12:26] "GET /lidar HTTP/1.1" 200 -
192.168.50.231 - - [10/Mar/2026 17:12:26] "GET /odometry HTTP/1.1" 200 -
192.168.50.231 - - [10/Mar/2026 17:12:26] "GET /odometry HTTP/1.1" 200 -
192.168.50.231 - - [10/Mar/2026 17:12:26] "GET /lidar HTTP/1.1" 200 -
192.168.50.231 - - [10/Mar/2026 17:12:26] "GET /odometry HTTP/1.1" 200 -
192.168.50.231 - - [10/Mar/2026 17:12:26] "GET /lidar HTTP/1.1" 200 -
```

3. [LIMO] Run ‘limo_server_lidar.py’

Demonstration (2/4)

```
(gpu) eunjin@DESKTOP-CJ4J7QB:~/ros2_ws/IETF125-260310/SDV_Robocar$ python k8s_server.py
 * Serving Flask app 'k8s_server'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI
server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:8080
 * Running on http://115.145.178.171:8080
Press CTRL+C to quit
[2026-03-10_18-12-27] 이미지 저장 완료 → logs/images/2026-03-10_18-12-27.jpg
[2026-03-10_18-12-27] 수신 완료 | 객체 수: 7
127.0.0.1 - - [10/Mar/2026 18:12:27] "POST /inference HTTP/1.1" 200 -
[2026-03-10_18-12-27] 이미지 저장 완료 → logs/images/2026-03-10_18-12-27.jpg
[2026-03-10_18-12-27] 수신 완료 | 객체 수: 6
127.0.0.1 - - [10/Mar/2026 18:12:27] "POST /inference HTTP/1.1" 200 -
[2026-03-10_18-12-28] 이미지 저장 완료 → logs/images/2026-03-10_18-12-28.jpg
[2026-03-10_18-12-28] 수신 완료 | 객체 수: 4
```

4. Run ‘k8s_server.py’

```
(gpu) eunjin@DESKTOP-CJ4J7QB:~/ros2_ws/IETF125-260310/SDV_Robocar$ python intent_server.py
 * Serving Flask app 'intent_server'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI
server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://115.145.178.171:5000
Press CTRL+C to quit
```

5. Run ‘intent_server.py’

Demonstration (3/4)

```
(gpu) eunjin@DESKTOP-CJ4J7QB:~/ros2_ws/IETF125-260310/SDV_Robocar$ python3 edge_control.py
[Odom] 시작
[LiDAR] 시작
[Infer] 시작
[Sender] 시작
[Capture] 시작
[Sender] 전송 성공: 2026-03-10_18-12-27
[Sender] 전송 성공: 2026-03-10_18-12-27
[Robocar] distance -> 2.07 m 전송 OK
[Robocar] distance -> 1.98 m 전송 OK
[Sender] 전송 성공: 2026-03-10_18-12-28
[Robocar] distance -> 1.45 m 전송 OK
[Robocar] distance -> 1.89 m 전송 OK
[Sender] 전송 성공: 2026-03-10_18-12-39
```

6-1. Run ‘edge_control.py’: pass

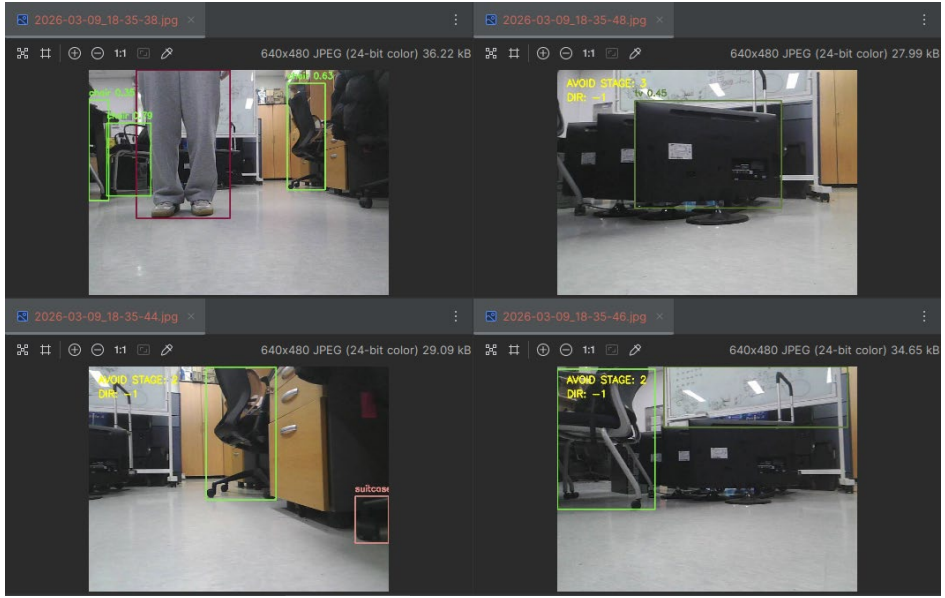
```
(gpu) eunjin@DESKTOP-CJ4J7QB:~/ros2_ws/IETF125-260310/SDV_Robocar$ python3 edge_control.py
[Odom] 시작
[LiDAR] 시작
[Infer] 시작
[Sender] 시작
[Capture] 시작
[Policy] ingress-action=drop -> distance 전송 차단 (1.74 m)
[Policy] ingress-action=drop -> distance 전송 차단 (2.12 m)
[Sender] 전송 성공: 2026-03-10_18-09-12
[Policy] ingress-action=drop -> distance 전송 차단 (1.84 m)
[Policy] ingress-action=drop -> distance 전송 차단 (2.12 m)
[Policy] ingress-action=drop -> distance 전송 차단 (2.11 m)
[Policy] ingress-action=drop -> distance 전송 차단 (1.84 m)
[Policy] ingress-action=drop -> distance 전송 차단 (1.84 m)
[Policy] ingress-action=drop -> distance 전송 차단 (1.84 m)
[Policy] ingress-action=drop -> distance 전송 차단 (2.11 m)
```

6-2. Run ‘edge_control.py’: drop

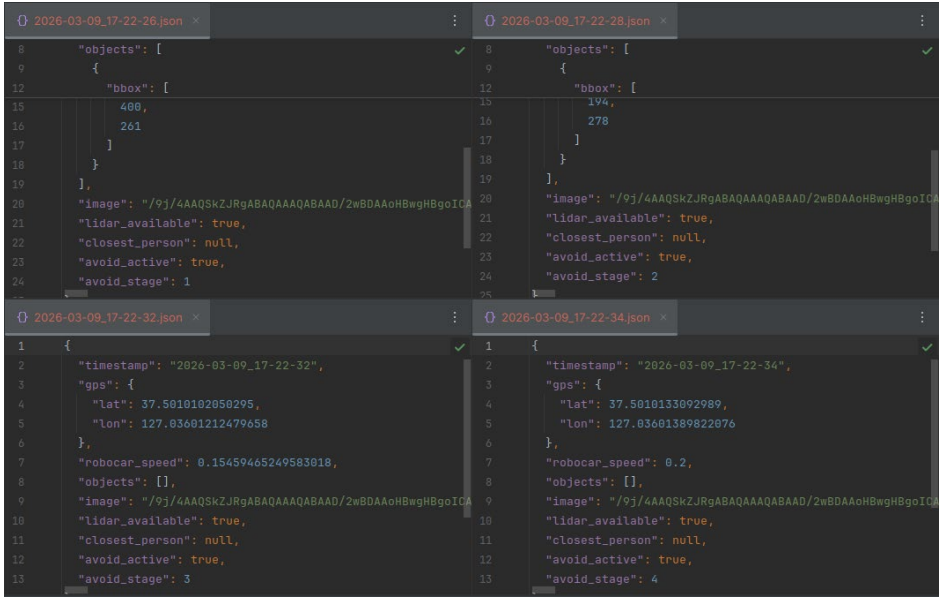
```
agilex@agikex-NUC12WSKi7: ~/SDV_Robocar
agilex@agikex-NUC12...   agilex@agikex-NUC12...   agilex@agikex-NUC12...   agilex@agikex-NUC12...
agilex@agikex-NUC12WSKi7:~/SDV_Robocar$ python3 imo_control.py
 * Serving Flask app 'imo_control'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI s
erver instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:8001
 * Running on http://192.168.50.165:8001
Press CTRL+C to quit
[INFO] [1773048935.384986355] [imo_limo_controller]: [IMO] Controller started | default_speed=0.2
192.168.50.17 - - [09/Mar/2026 17:35:40] "POST /control/cmd_vel HTTP/1.1" 200 -
192.168.50.17 - - [09/Mar/2026 17:35:40] "POST /control/cmd_vel HTTP/1.1" 200 -
192.168.50.17 - - [09/Mar/2026 17:35:41] "POST /control/cmd_vel HTTP/1.1" 200 -
192.168.50.17 - - [09/Mar/2026 17:35:41] "POST /control/cmd_vel HTTP/1.1" 200 -
192.168.50.17 - - [09/Mar/2026 17:35:41] "POST /control/cmd_vel HTTP/1.1" 200 -
192.168.50.17 - - [09/Mar/2026 17:35:41] "POST /control/cmd_vel HTTP/1.1" 200 -
192.168.50.17 - - [09/Mar/2026 17:35:41] "POST /control/cmd_vel HTTP/1.1" 200 -
192.168.50.17 - - [09/Mar/2026 17:35:41] "POST /control/cmd_vel HTTP/1.1" 200 -
192.168.50.17 - - [09/Mar/2026 17:35:41] "POST /control/cmd_vel HTTP/1.1" 200 -
192.168.50.17 - - [09/Mar/2026 17:35:41] "POST /control/cmd_vel HTTP/1.1" 200 -
192.168.50.17 - - [09/Mar/2026 17:35:41] "POST /control/cmd_vel HTTP/1.1" 200 -
```

7. Run ‘limo_control.py’

Demonstration (4/4)



Result 1: YOLO Detection & Distance Measurement



Result 2: Logs

What we learned

- We implemented a Real-time Perception-to-Avoidance Pipeline in the I2ICF Framework.
  - Sending a User Intent to Activate or Deactivate Stop/Detour Function in a Robot Car.
  - Measuring Distance with the Robot Car's Camera and LiDAR.
  - Auto-Avoiding an Obstacle for Safety.

- Intent-Based Driving
  - The robot car automatically plans a moving path when an obstacle is nearby.

Next Step

- **Design & Implementation of Intent Translator and Policy Translator for Intelligent Moving Objects (e.g., Robot Car and Humanoid).**

- **IETF-126 Hackathon Project**
  - Extension of I2ICF Testbed
    - Collision Avoidance between Robot Cards using Vehicle-to-Vehicle (V2V) Communication
  - YANG Data Models for I2ICF Moving Object Management
    - https://datatracker.ietf.org/doc/draft-jeong-opsawg-i2icf-framework/

Open-Source Project for I2ICF

[URL] https://github.com/jaehoonpauljeong/I2ICF/tree/main/IETF-125

jaehoonpauljeong / I2ICF

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Files

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  - README.md
  - Readme.txt
  - edge_control.py
  - images.png
  - imo_control.py
  - imo_server_lidar.py
  - intent_server.py
  - json.png
  - k8s_server.py
  - received_policy.yaml
- Side-Meeting
- README.md
- Readme

I2ICF / IETF-125 /

Add file

cowltnr 260310 update

916e584 · 3 minutes ago History

Name	Last commit message	Last commit date
detector	260310 update	3 hours ago
logs	260310 update	3 hours ago
README.md	260310 update	3 minutes ago
Readme.txt	Create Readme.txt	4 hours ago
edge_control.py	260310 update	3 hours ago
images.png	260310 update	3 hours ago
imo_control.py	260310 update	3 hours ago
imo_server_lidar.py	260310 update	3 hours ago
intent_server.py	260310 update	3 hours ago
json.png	260310 update	3 hours ago
k8s_server.py	260310 update	3 hours ago
received_policy.yaml	260310 update	3 hours ago

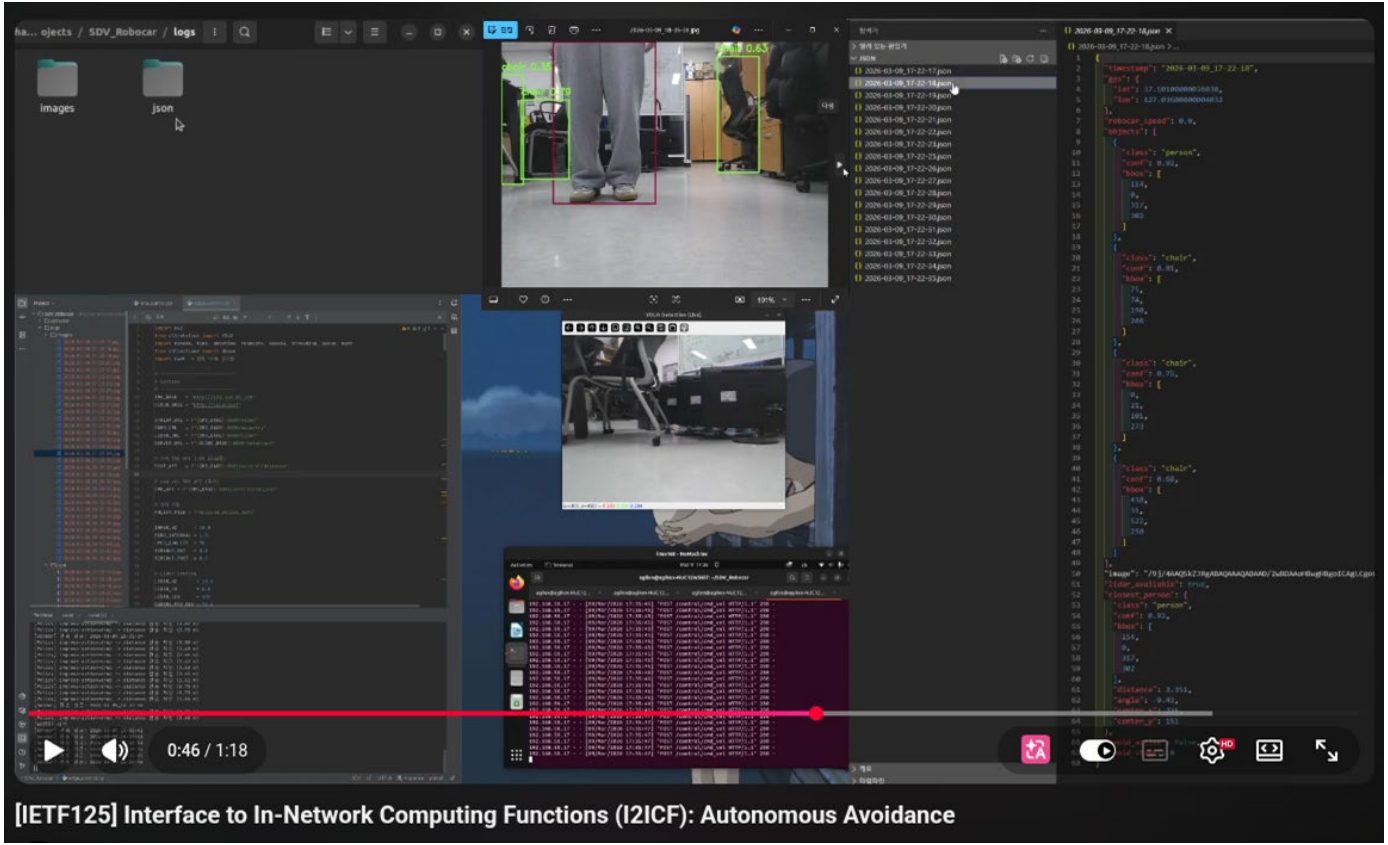
README.md

Main steps

LIMO

Demonstration Video Clip for I2ICF

[URL] https://youtu.be/589GNGIX3fk?si=oZB06Nk9Y8ozSOLz



I2ICF Hackathon Team

- **Professors:**
  - Jaehoon Paul Jeong (SKKU)
  - Yong-Geun Hong (DJU)
  - Joo-Sang Youn (DEU)
  - Pyung Soo Kim (TU Korea)
- **Researchers:**
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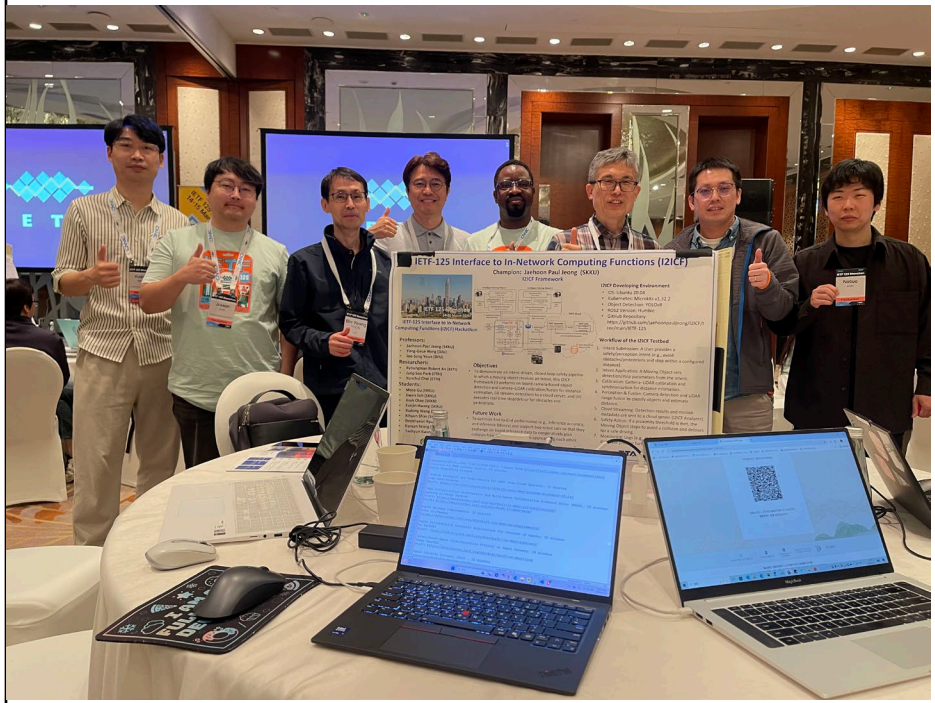
  - Kehan Yao (China Mobile)
  - Byoungman Robert An (KETI)
  - Jung-Soo Park (ETRI)
  - Yunchul Choi (ETRI)
  - Bin Yeong Yoon (ETRI)
- **Students:**
  - Mose Gu (SKKU) , Jiwon Suh (SKKU), Jisuk Chae (SKKU)
  - Eunjin Hwang (SKKU), Xudong Wang (SKKU)
  - Kihyun Shim (SKKU)
  - Deokhyeon Ryu (DJU), Eunsan Seong (DEU)
  - Taehyun Kwan (DEU), Nobuo Aoki (Independent)

Hackathon Team Photo

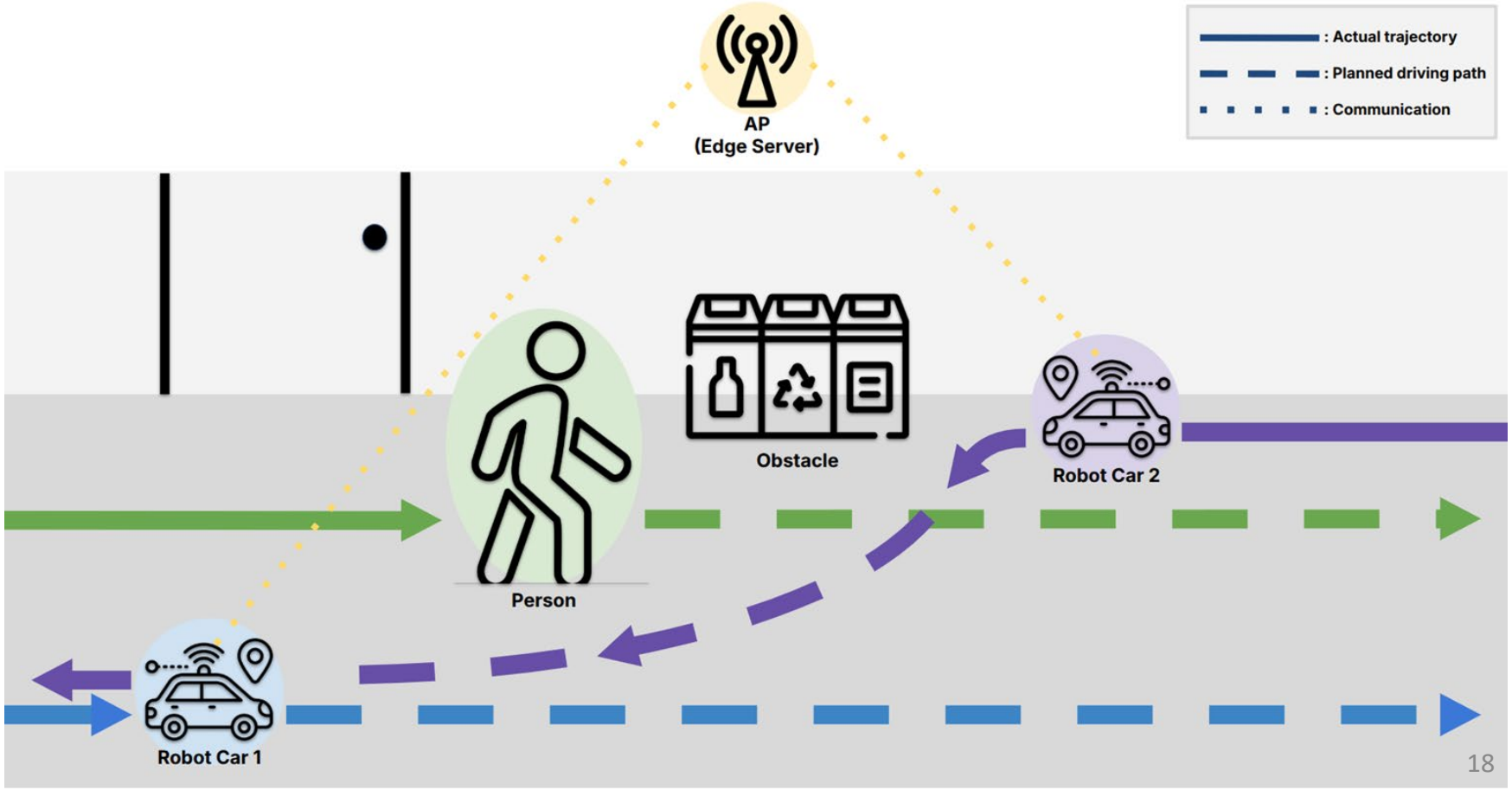


Appendix

Differences from the previous work

- The Previous Goal
  - To perform onboard camera-based object detection and camera-LiDAR calibration/fusion for distance estimation.
  - To let the moving object deliver the detection results to a cloud server to perform real-time stopping to avoid the collision with obstacles.
  - Camera-LiDAR data is calibrated and fused; collision control is performed via the powertrain ECU; detection results are transferred to a cloud server.
- The Present Goal
  - To have a moving object (i.e., Robot Car) receive an intent and autonomously drive.
  - The robot automatically plans a detour when an object gets too close.

Final Goal Scenario Illustration



Thank you!